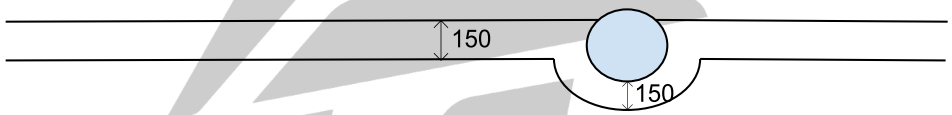
Inter IIT Tech Meet 14.0
Eternal
Answers to Queries

1. **Clarification Regarding Minimum and Maximum Distance from Shelf**

    - The robot's wheels which are facing the rack should not cross the minimum and maximum distance lines.

2. **Clarification Regarding Obstacles**

    - The robot's wheels which are facing the obstacle should always be at a distance of 15 cm.
    - The maximum and minimum distance along the obstacle will be as illustrated below:



    - The obstacle will not be covering any QR Code.
    - The position of the obstacle will not be fixed as per the arena layout; it may change during runs.

3. **Use of Edge Server with Separate Power Supply**

    - YES, edge computing is permitted.
    - External laptop/desktop/edge server connected via WiFi for image processing, path planning optimization or data logging are allowed.
    - Edge server can use venue AC power or your own UPS/generator

4. ISO and IEC Standards Compliance

- Compliance is required, but full document purchase is not mandatory.
- Critical requirements include:
  - Emergency stop: < 500ms
  - Active obstacle detection system
  - Maximum speed in shared spaces: 1.0 m/s
  - Reduced speed near obstacles: 0.3 m/s
  - Visual/audio warnings when moving
  - Robot must fail-safe if WiFi connection is lost (e.g., stop and wait, or return to start position)
- Teams do not need to purchase the full ISO/IEC documents (priced at $200+). Follow the summarized safety guidelines provided.

5. Clarification About Markings

- The markings will be present only for evaluation purposes for the judges.
- They will indicate limits such as the minimum and maximum distance from the shelf.
- The color and material of these markings are not significant to be known.

6. QR Code Version Details

- Version: QR Code Version 2 (25×25 modules)
- Data capacity: Up to 20 alphanumeric characters (sufficient for R03_S2_ITM451)
- Physical size: 50mm × 50mm
- Error correction level: M (15% recovery capacity)
- Print: Black on white background, matte finish

7. Vertical Positioning Accuracy

- Vertical positioning accuracy (±2cm) refers to camera elevation control.

- When commanding “move camera to height 1.0 m”, the actual height should be between 0.98 m and 1.02 m.
- Measured from ground to the center of the camera lens.
- Minimum camera height: 20cm (must clear ground obstacles, cable management)
- Maximum extended height: 1.8m
- This ensures repeatable scanning of shelf levels across multiple runs

8. Wi-Fi and Network Connectivity

- Teams may use their own network.
- Venue Wi-Fi is not guaranteed to be available or reliable.
- Bringing your own router or hotspot is recommended.
- If you require venue AC power for your network setup, notify the organizers 48 hours in advance.
- Network Requirements: Use WPA2/WPA3 encryption; no specific protocols mandated.
- Ensure an offline fallback mode in case connectivity fails during demonstration.

9. Bonus Points for Early Completion

- YES, a time-based bonus available.
- However, accuracy and safety take precedence over speed.
- Any failed QR scans, collisions, or safety violations will negate time bonuses.

10. Robot Navigation and Rack Scanning

- Autonomous operation is preferred.
- The robot may be instructed to move to a particular rack or scan all racks autonomously and return to the end position.

11. Scanning Repeatability

- Scanning repeatability refers to the robot's ability to produce consistent results across multiple runs.
- When scanning the same rack 10-20 times, at least 95% of the scans should yield:
    - The same QR codes detected.
    - Comparable image quality.
    - Similar robot positioning.

12. Manual Mapping

- YES, teleoperated mapping is allowed during the mapping phase.
- You may use a joystick, keyboard, or tablet interface to drive the robot.
- Use joystick, keyboard, or tablet interface to drive robot
- SLAM runs autonomously during teleoperation.
- After mapping, the robot must operate fully autonomously for the scanning mission.
- Scoring: No penalty for teleoperated mapping.

13. Clarification Regarding QR Code Scanning Alignment

- While scanning the QR code, the robot must remain within the length of the shelf.
- The shelf length is 910 cm; therefore, the side of the robot that is parallel to the shelf and the scanning position must lie entirely within these boundary lines.
- The robot should not scan the QR code at an angle; alignment must be parallel to the shelf for accurate detection.
- Markings will be drawn on the arena floor to indicate these limits.